PLANET • CLIMATE RESILIENCE

Flood-Ready Township

Know before the water reaches your street.

On 12 Nov 2024, Bukit Raja's roads flooded and no street got its own warning. We built one.

A web app that warns residents of Bandar Bukit Raja street by street before a flash flood, and gives SDP a record of where and when floods happen.



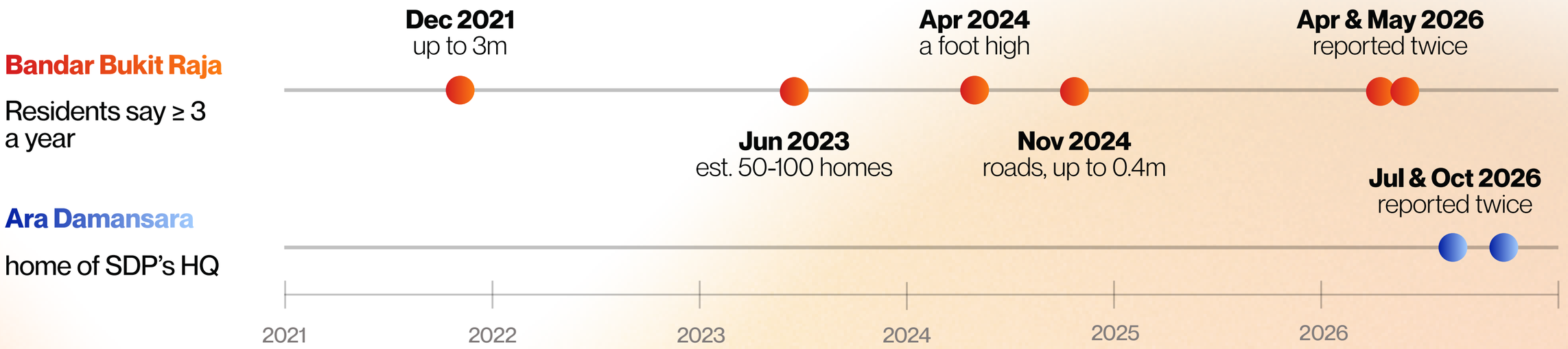
Sime Darby Property University Hackathon 2026

Team bogosort

PROBLEM

Flash floods keep coming back to SDP townships

Each dot is one flood reported by the news or Bomba. Gaps mean no report was found, not no flood.



> "How do we help them keep living sustainably long after the keys are handed over?"
>
> SDP, Township of Tomorrow brief

> Flooding is "expected to remain a significant risk across certain townships" to 2050.
>
> SDP Integrated Annual Report 2025, p.125

Sources: FMT 24 May 2024 · Sinar Harian 25 May 2024, 6 Jun 2023 · RTM 12 Nov 2024, 20 Apr and 31 Jul 2026 · Bernama 8 May 2026 · The Star 17 Oct 2026

PROBLEM: WHY IT MATTERS TO SDP

Flood is on SDP's risk register, but its cost hasn't been published

"Implement flood monitoring and emergency response protocols in high-risk townships"

Principal risk: Climate Risk · IAR 2025 p.137

"Establish climate adaptation and resilience plans for high-risk assets"

SSR2030 target, Climate Resilience pillar · SDP hackathon briefing

"Quantitative financial effects have not been disclosed"

IAR 2025 p.203 · IFRS S2 reporting from 2027 (AGM p.7)

78% of Malaysians surveyed weigh climate risk when buying a home

PropertyGuru consumer sentiment study, H1 2026

BUKIT RAJA FLOOD LOSS, 2024

ESTIMATE

Low • RM 1.3m

Base • RM 6.3m (SDP RM4.3m)

High • RM 22.8m



Residents SDP

Counted: residents' home contents and cars; SDP's drainage works, sales discounts and complaint handling. Inputs are our assumptions. SDP's base cost is 0.9% of FY2024 profit.

Estimate: our model, inputs labelled in our project repo · FY2024 PATAMI RM502.2m (AGM p.19) · PropertyGuru H1 2026

TARGET USER

Who it serves: residents, then SDP's teams

PRIMARY USER

Residents of Bukit Raja Phase 1–2

"Will my street flood, when, and what should I do?"

- 583 residents, 48 associations signed a memorandum (May 2024)
- Residents report ≥ 3 flash floods a year since Dec 2021
- Water inside kitchens and living rooms



SDP TOWNSHIP OPS

Act before the calls come in

"Where is risk rising, and what do I tell residents?"

Today, public gauges show rivers, not streets.

SDP PLANNING, RISK

Fix the right drain first

"Which streets flood most, and what does it cost us?"

Today, a desktop screen of 7 townships; cost not yet quantified.

NPS 88%

SDP's reported Net Promoter Score is measured at **key handover**, not after residents live through a flood.

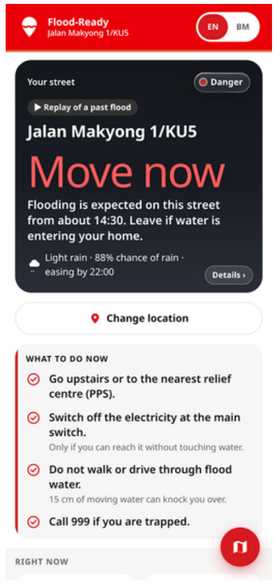
Sources: FMT 24 May 2024 · Sinar Harian 25 May 2024 · SDP IAR 2025 p.147 (NPS), p.203 (seven-township pilot)

SOLUTION

Flood-Ready Township: a warning for your street, and evidence for SDP

Resident App · installable, BM / EN

Open app ↗





- One word says what to do: Safe, Be alert, Get ready, Move now
- Push alerts: Warning and Danger always, with a time window; Watch at most every 12 h
- Flood map and “Use my location”
- “Is it flooding here?”: Yes, No or Not sure

SDP admin app · township ops

Open admin ↗

- Live map of all 943 Bukit Raja streets
- Review queue of residents’ answers and flood reports
- Statistics: alerts by week and by street, station health
- Automatic monthly report
- Drainage suggestions from repeat alerts and confirmed floods

Real app screens: replay of the 12 Nov 2024 flood at 14:00 · Map: © OpenStreetMap contributors · JPS Public Infobanjir · MET Malaysia · Open-Meteo · FABDEM (Univ. of Bristol, CC BY-NC-SA) · ESA WorldCover

INNOVATION

What’s new: river gauges become street alerts, checked by residents

	Today	Flood-Ready Township
Where	Rain gauges up to 9 km apart	Each of 943 streets: its own river and ≥ 2 rain gauges
When	The water level right now	Level and reason every 15 min; a time window from Warning up
Ahead	District weather warnings	6–24 h township outlook from forecasts and MET warnings
Trust	A dead gauge stays published	Gauges checked against each other; we found Kg Perepat frozen since Jul 2024
Proof	No record of which street flooded	Residents answer Yes, No or Not sure; each street builds a flood history
Action	Drains argued from complaints	Drain suggestions from repeat alerts and confirmed floods

Sources: JPS Public Infobanjir · our QC and gauge-spacing findings (docs/findings.md A3c, D1)

HOW IT WORKS · RUNS EVERY 15 MINUTES

From public data to your phone in five steps



STEP 1

Collect

JPS river and rain gauges, tide, MET warnings, forecast

STEP 2

Check

Set aside dead, stuck or spiking gauges

STEP 3

Decide

JPS-based rules set the level; ML shows a % chance

STEP 4

Locate

Each of 943 streets gets that level or lower, by how flood-prone it is

STEP 5

Alert

Push to phones (Watch at most every 12 h); live map for SDP

Watch trigger

Rain ≥ 31 mm in 1 h or ≥ 40 mm in 2–4 h, or river at alert mark

Warning trigger

Rain over 60 mm in 1 h or ≥ 60 mm in 2–4 h, or river at warning mark

Danger trigger

River at its danger mark

ML · shown as a %, not alerting

Chance the river reaches its warning mark within 2 h. May raise Watch only, once it passes our test (see Feasibility)

Then it feeds back

Residents answer "Is it flooding here?" → Each street builds a flood history → SDP gets drainage suggestions

River marks are JPS's own; rain figures follow JPS rainfall categories (40 mm in 2–4 h is our starting value). MET warnings and forecast drive only the 6–24 h outlook.

IMPACT

Had it run in 2024: about 8% of the loss avoided. The rest needs drains

8% ESTIMATE

Base case: **RM0.49m** of an estimated **RM6.3m**

Low 1% · High 18% · if half of homes act

Where it comes from

- Cars moved and belongings lifted: in 4,468 German flood damage cases, warnings ≥ 1 h ahead cut contents damage ~19%
- Drainage money aimed at the streets that flood
- Fewer hours spent on complaints
- Costs ~ RM20k/yr per township: ~11× back for SDP (base case)

WE WILL MEASURE IN THE PILOT

Share of resident-confirmed floods warned ≥ 30 min ahead · false Warnings per month · households subscribed

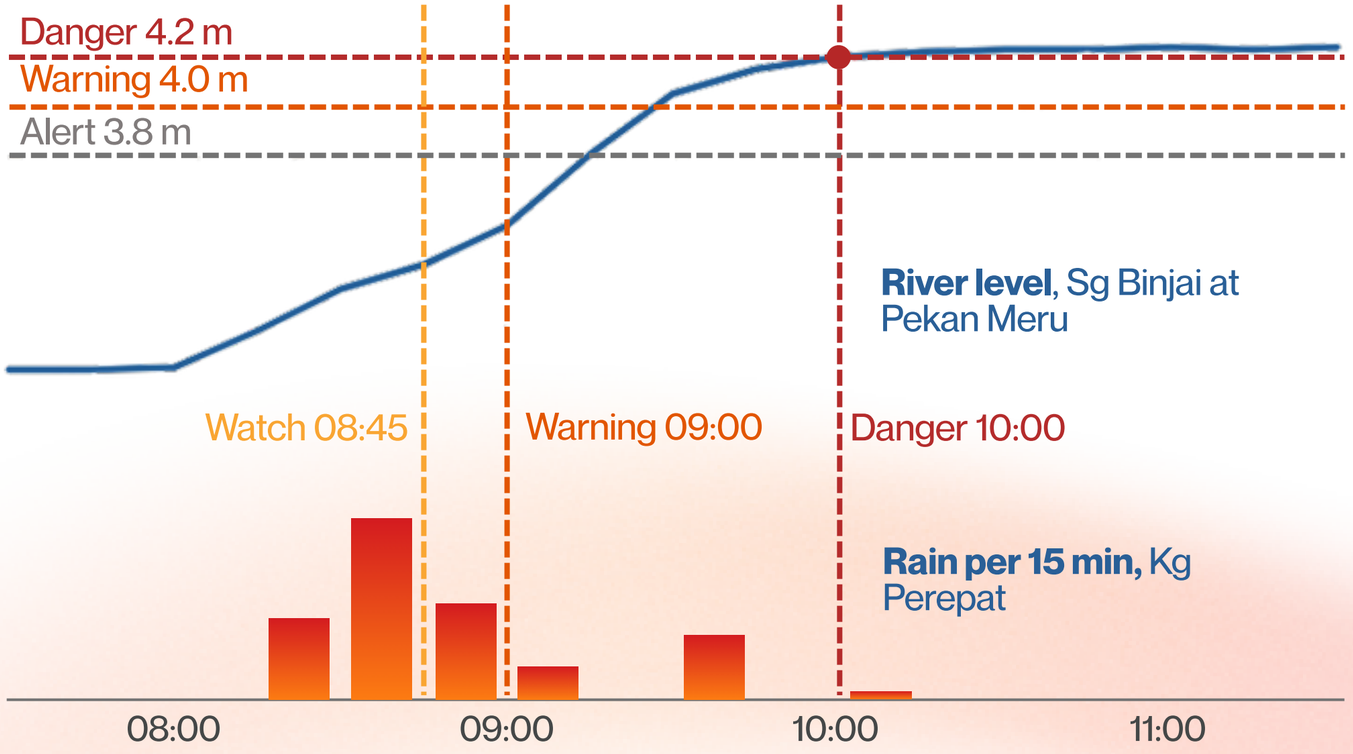
WHAT IT CAN'T FIX

It drains no water · it's only as good as rain gauges up to 9 km apart · JPS, MET and NADMA stay the official warning

Estimate: inputs labelled in our project repo · Kreibich, Hudson, Merz (EGU 2022): warnings ≥ 1 h ahead, people knew what to do · Costs: pilot budget [sheet link]

FEASIBILITY: TESTED ON REAL DATA

Replay of 12 Nov 2024: an hour of warning before the river hit danger



Streets alerted

- **320** at Watch, 08:45
- **227** at Warning, 09:00
- **323** at Danger, 10:00

Tested on 890 river rises

Times an inland river crossed its warning mark, 2024–26. JPS rules warned before **795,** a median **90 min** ahead; with ML switched on, 135 min.

On 49 rivers held out from design, ML added only 15 min (our pre-set bar: 30), so ML only shows a %; the rules raise alerts.

Data: JPS Public Infobanjir 15-min readings · replay with models that never saw this period · docs/evaluation-report.md

IMPLEMENTATION

2027: three SDP townships live, with sensors in the drains

PHASE 0 · NOW – DEC 2026

Hackathon MVP

- Open data, replay and live
- Resident and admin apps

Gate: SDP sponsor; legal OK on warning wording

PHASE 1 · JAN – JUN 2027

Bukit Raja pilot

- Residents join via their associations
- 6–10 LoRaWAN water sensors (~ USD 90–430 each) on the memorandum streets
- Live through the Apr–May storms

Gate: ≥ 70% of floods warned ≥ 30 min ahead

PHASE 2 · JUL – DEC 2027

Drains + 3 townships

- Sensors at ponds, flap gates and pumps
- Inside the SDP App and PACC
- Data shared with MBDK and JPS

Gate: results hold; one drain fix chosen by data

PHASE 3 · 2028 ONWARD

High-risk townships

- Pumps pre-started, with human approval
- Climate data for IFRS S2
- Land-bank flood screening

Review: yearly, same metrics

2027 target

3 SDP townships live · ≥ 20 water sensors in streets and drains · ≥ 70% of resident-confirmed floods warned ≥ 30 min ahead

Gate = what must be true before the next phase starts · Partners: sensor maker + IoT network, drainage operator, none signed yet · Budget: pilot ~ RM37k in year 1, then ~ RM20k per township a year (list prices; estimates labelled): [sheet link] · Targets are our estimates